

Classification

K Arnold

Objectives

- Apply the `tidymodels` pipeline to a classification task
- Identify when validation is necessary to believe an outcome
- Apply the concepts of sensitivity and specificity

(punted to next time:)

- Identify corresponding elements between R and Python data wrangling and modeling workflows

Outline

- Problem introduction
 - Data wrangling in R
- Classification workflow
 - Models: decision tree, logistic regression
 - Model outputs: scores and decisions
 - Model metrics: accuracy, sensitivity, specificity
 - Validation

Logistics notes

- Discussion on fairness definitions posted, due next Tuesday.
- Your **project** should be
 - **Interesting**: not every project will go into depth in every aspect (e.g., some won't have much data wrangling), but all projects should be interesting in *some* aspect.
 - **Your own**: examples abound on the Internet. Following a tutorial is a very boring project. Adapting its approach to a new dataset or question? Interesting.

Setup

```
library(tidyverse)
library(tidymodels)
library(ggribes)
```

(r)

Can a blood test diagnose autism?

We'll use an example from <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005385>

```
data_filename <- "data/autism.csv" (r)
if (!file.exists(data_filename)) {
  dir.create("data")
  download.file("https://doi.org/10.1371/journal.pcbi.1005385.s001", data_filename)
}
```

```
col_names <- names(read_csv(data_filename, n_max = 1, col_types = cols(.default = col_character())))
autism <- read_csv(data_filename, skip = 2, col_names = col_names, col_types = cols(
  .default = col_double(),
  Group = col_character()
)) %>% mutate(
  Group = as_factor(Group)
)
```

We have 3 kinds of data about 206 children:

1. The outcome (**Group**): ASD (diagnosed with ASD), SIB (sibling not diagnosed with ASD), and NEU (age-matched neurotypical children, for control)

```
autism %>% group_by(Group) %>% summarize(n = n()) %>% kable() (r)
```

Group	n
ASD	83
NEU	76
SIB	47

1. The outcome (**Group**): ASD, SIB, NEU

2. Concentrations of various metabolites in a blood sample:

```
autism %>% select(-1, -last_col())
```

(r)

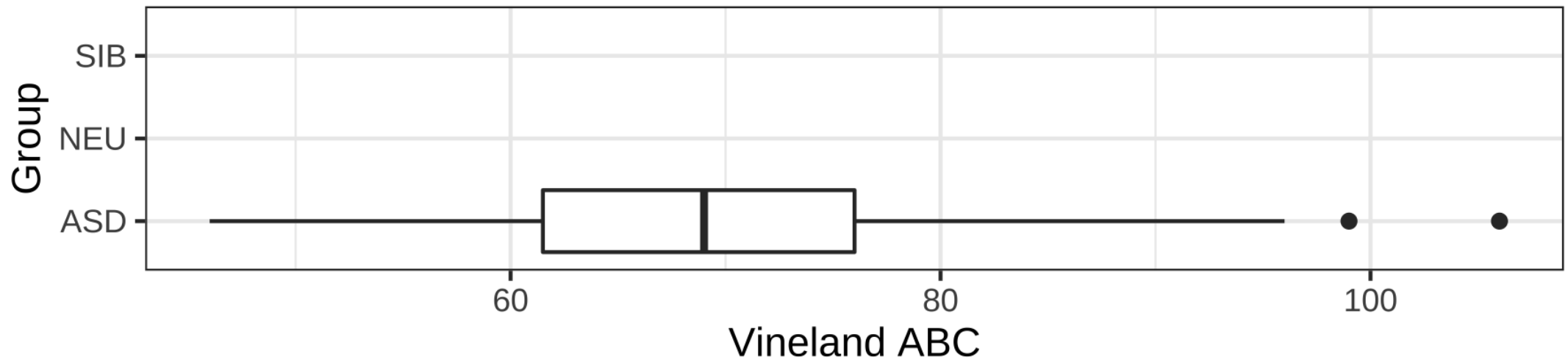
```
## # A tibble: 206 x 24
##   Methion.    SAM    SAH `SAM/SAH` `% DNA methylat... `8-OHG` Adenosine Homocysteine Cysteine
##   <dbl> <dbl> <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1    17.3  56.2  15.2     3.69     3.35    0.055    0.103     4.48    215.
## 2    14.9  37.2   7.58     4.91     3.04    0.045    0.055     4.31    163.
## 3    15.9  37.9   9.87     3.84     2.81    0.058    0.071     6.21    159.
## 4    18.7  79.2  24.5     3.23     4.24    0.085    0.079     7.15    226.
## 5    21.5  77.6  19.2     4.04     3.49    0.041    0.058     3.82    201.
## 6    18.1  67.6  12.8     5.30     3.01    0.156    0.091     4.03    205.
## # ... with 200 more rows, and 15 more variables: `Glu.-Cys.` <dbl>, `Cys.-Gly.` <dbl>, tGSH <dbl>,
## #   fGSH <dbl>, GSSG <dbl>, `fGSH/GSSG` <dbl>, `tGSH/GSSG` <dbl>, Chlorotyrosine <dbl>,
## #   Nitrotyrosine <dbl>, Tyrosine <dbl>, Tryptophane <dbl>, fCystine <dbl>, fCysteine <dbl>,
## #   `fCystine/fCysteine` <dbl>, `% oxidized` <dbl>
```


1. The outcome (**Group**): ASD, SIB, NEU
2. Concentrations of various metabolites in a blood sample
3. For the ASD children only, a measure of life skills ("Vineland ABC")

```
autism %>%  
  ggplot(aes(x = `Vineland ABC`, y = Group)) + geom_boxplot()
```

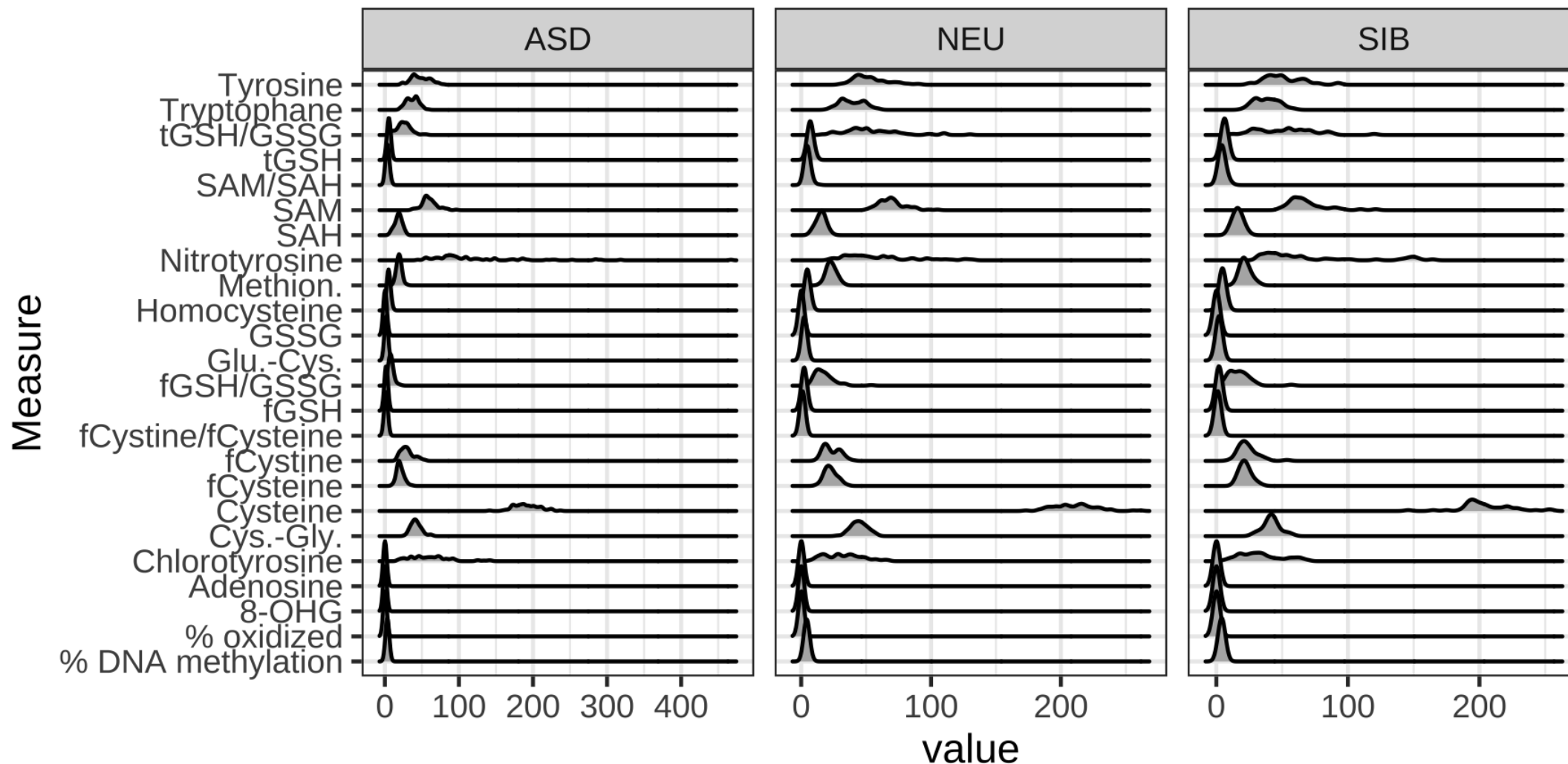
(r)

Warning: Removed 159 rows containing non-finite values (stat_boxplot).



Exploratory Data Analysis (EDA)

What do these metabolites look like?



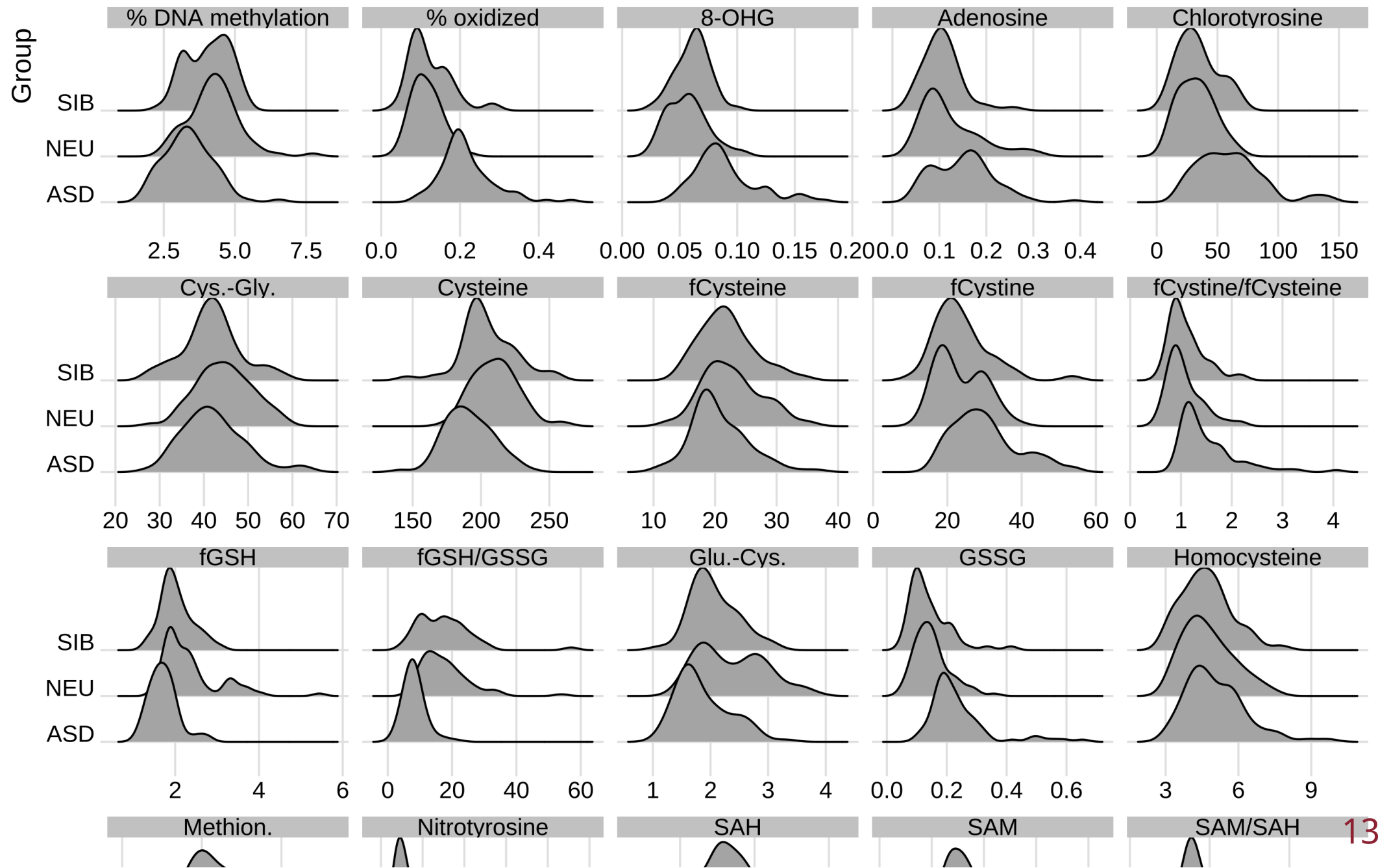
code for the previous plot:

```
autism %>%  
  select(-`Vineland ABC`) %>%  
  pivot_longer(-Group, names_to = "Measure") %>%  
  ggplot(aes(x = value, y = Measure)) +  
  geom_density_ridges() +  
  facet_wrap(vars(Group), scales = "free_x")
```

(r)

EDA

Better question: **Can these metabolites help us distinguish autism?**



code for previous plot:

```
autism %>%  
  select(-`Vineland ABC`) %>%  
  pivot_longer(-Group, names_to = "Measure") %>%  
  ggplot(aes(x = value, y = Group)) +  
  geom_density_ridges() +  
  facet_wrap(vars(Measure), scales = "free_x") +  
  theme_ridges()
```

(r)

Can we predict ASD vs non-ASD from metabolites?

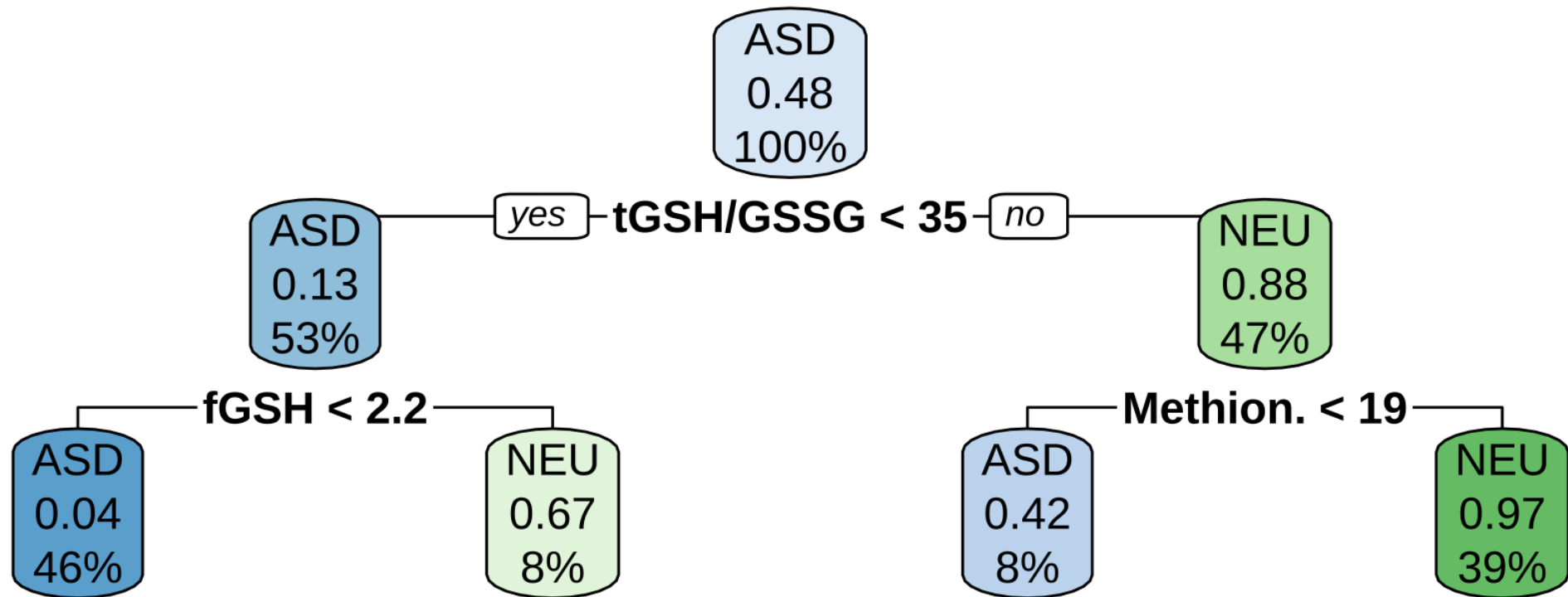
- Let's start by (1) ignoring the behavior scores (that's an *outcome*) and comparing just ASD and NEU.
- We need to drop SIB... and tell the model that we don't actually care about it.

```
data <-  
  autism %>%  
  select(-`Vineland ABC`) %>%  
  filter(Group != "SIB") %>%  
  mutate(Group = factor(Group))
```

(r)

Decision Tree Classification

```
spec <- workflow() %>% add_recipe(  
  recipe(Group ~ ., data = data)) %>%  
  add_model(decision_tree(mode = "classification") %>% set_engine("rpart"))  
model <- spec %>% fit(data) (r)
```



What do the *predictions* look like?

```
model %>% predict(data, type = "prob")
```

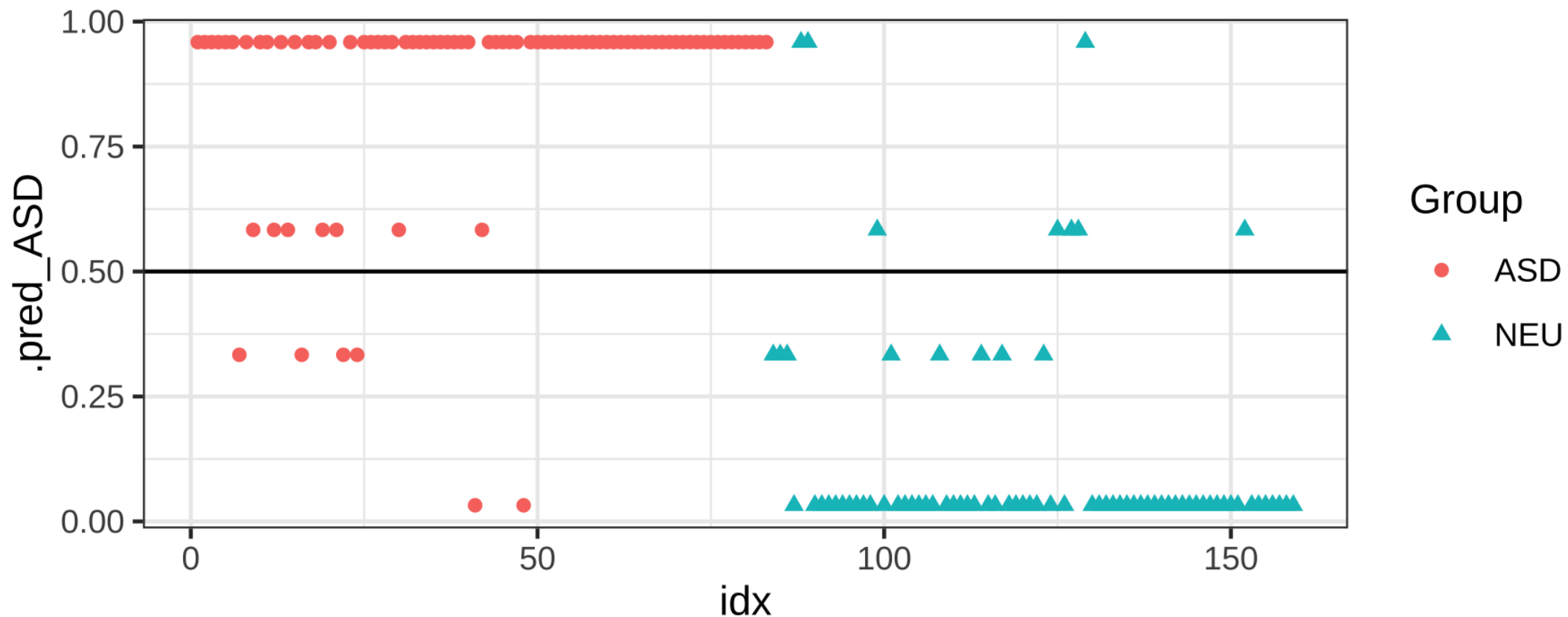
(r)

```
## # A tibble: 159 x 2
##   .pred_ASD .pred_NEU
##   <dbl>     <dbl>
## 1     0.959     0.0411
## 2     0.959     0.0411
## 3     0.959     0.0411
## 4     0.959     0.0411
## 5     0.959     0.0411
## 6     0.959     0.0411
## # ... with 153 more rows
```

Were those predictions good?

```
model %>%  
  predict(data, type = "prob") %>%  
  bind_cols(data) %>%  
  mutate(idx = row_number()) %>%  
  ggplot(aes(x = idx, y = .pred_ASD, color = Group, shape = Group)) +  
    geom_hline(yintercept = .5) +  
    geom_point()
```

(r)



Quantifying that:

```
metrics <- yardstick::metric_set(accuracy, sensitivity, specificity) (r)
model %>%
  predict(data, type = "class") %>%
  bind_cols(data) %>%
  metrics(truth = Group, estimate = .pred_class)
```

```
## # A tibble: 3 x 3
##   .metric .estimator .estimate
##   <chr>    <chr>      <dbl>
## 1 accuracy binary      0.912
## 2 sens     binary      0.928
## 3 spec     binary      0.895
```

Recall from Week 6...

	Seizure happened	No seizure happened
Seizure predicted	True positive	False positive (Type 1 error)
No seizure predicted	False negative (Type 2 error)	True negative

Recall from Week 6...

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Seizure predicted	True positive	False positive (Type 1 error)
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- **Accuracy** (% correct) = $(TP + TN) / (\# \text{ episodes})$
- **False negative** ("miss") **rate** = $FN / (\# \text{ actual seizures})$
- **False positive** ("false alarm") **rate** = $FP / (\# \text{ true non-seizures})$

Recall from Week 6...

	Seizure happened	No seizure happened
Seizure predicted	True positive	False positive (Type 1 error)
No seizure predicted	False negative (Type 2 error)	True negative

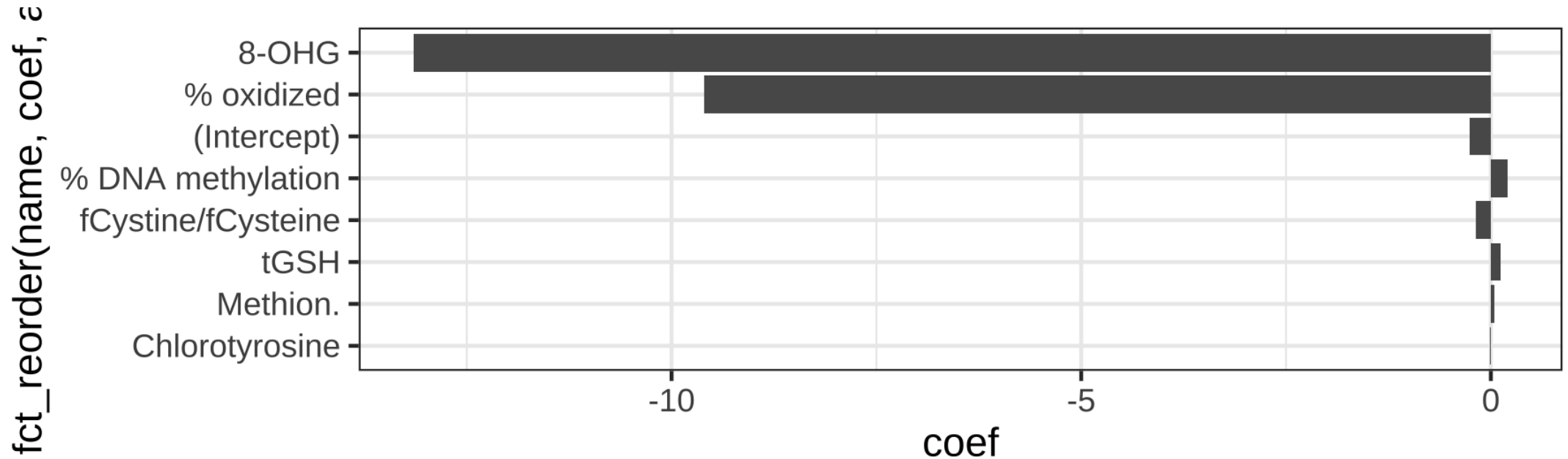
- **Accuracy** (% correct) = $(TP + TN) / (\# \text{ episodes})$
- **False negative** ("miss") **rate** = $FN / (\# \text{ actual seizures})$
- **False positive** ("false alarm") **rate** = $FP / (\# \text{ true non-seizures})$
- **Sensitivity** ("true positive rate") = $TP / (\# \text{ actual seizures})$
 - Sensitivity = $1 - \text{False negative rate}$
- **Specificity** ("true negative rate") = $TN / (\# \text{ actual seizures})$
 - Specificity = $1 - \text{False positive rate}$
- [Wikipedia article](#)

Logistic Regression

Logistic Regression

```
spec <- workflow() %>% add_recipe(  
  recipe(Group ~ ., data = data)) %>%  
  add_model(logistic_reg(penalty = .001) %>% set_engine("glmnet"))  
model <- spec %>% fit(data) (r)
```

```
model %>% pull_workflow_fit() %>% pluck('fit') %>% coef(s = .1) %>% as.matrix() %>% as_tibble(rownames = "name") %>%  
  rename(coef = 2) %>% filter(abs(coef) > .01) %>%  
  ggplot(aes(x = coef, y = fct_reorder(name, coef, abs))) + geom_col() (r)
```




```
model %>% predict(data, type = "prob")
```

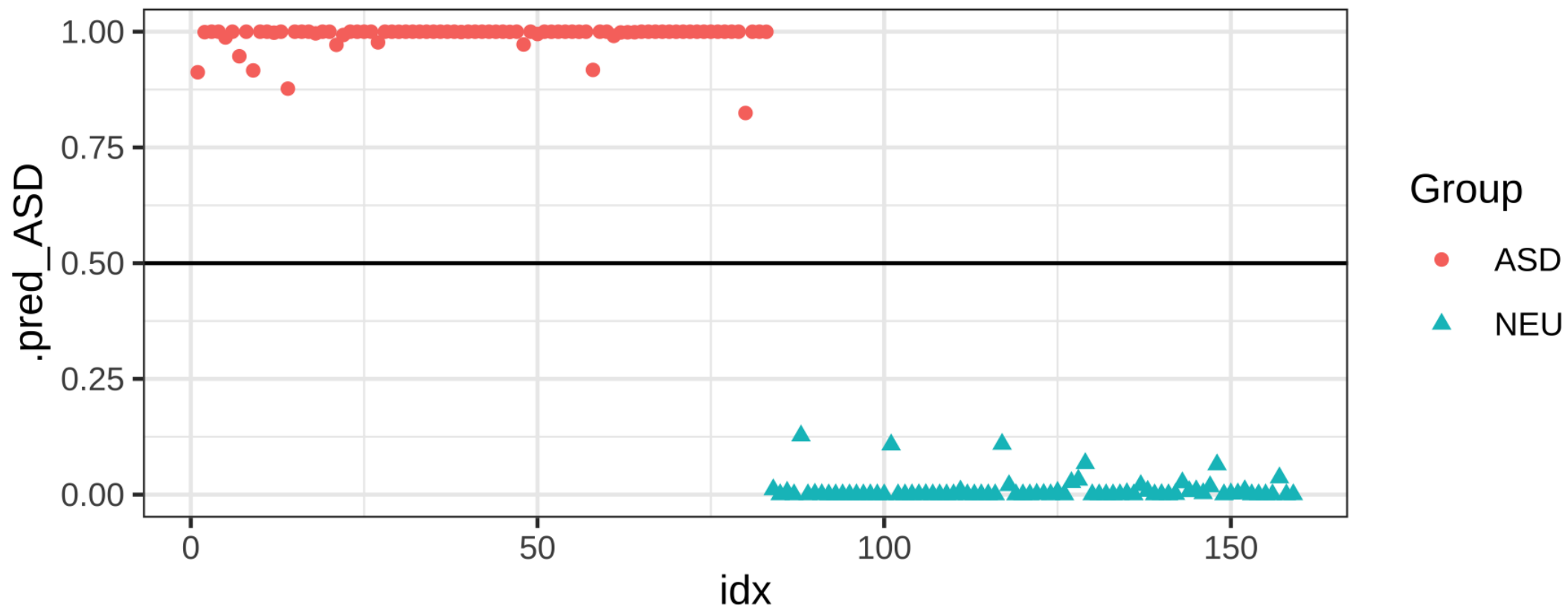
(r)

```
## # A tibble: 159 x 2
##   .pred_ASD .pred_NEU
##   <dbl>     <dbl>
## 1    0.912 0.0876
## 2    0.999 0.000964
## 3    1.00 0.0000000524
## 4    1.00 0.0000000369
## 5    0.988 0.0124
## 6    1.00 0.0000222
## # ... with 153 more rows
```

Are those predictions good?

```
model %>%  
  predict(data, type = "prob") %>%  
  bind_cols(data) %>%  
  mutate(idx = row_number()) %>%  
  ggplot(aes(x = idx, y = .pred_ASD, color = Group, shape = Group)) +  
    geom_hline(yintercept = .5) +  
    geom_point()
```

(r)



Quantifying that...

```
model %>%  
  predict(data, type = "class") %>%  
  bind_cols(data) %>%  
  metrics(truth = Group, estimate = .pred_class)
```

(r)

```
## # A tibble: 3 x 3  
##   .metric .estimator .estimate  
##   <chr>    <chr>         <dbl>  
## 1 accuracy binary         1  
## 2 sens     binary         1  
## 3 spec     binary         1
```

Cross validation!

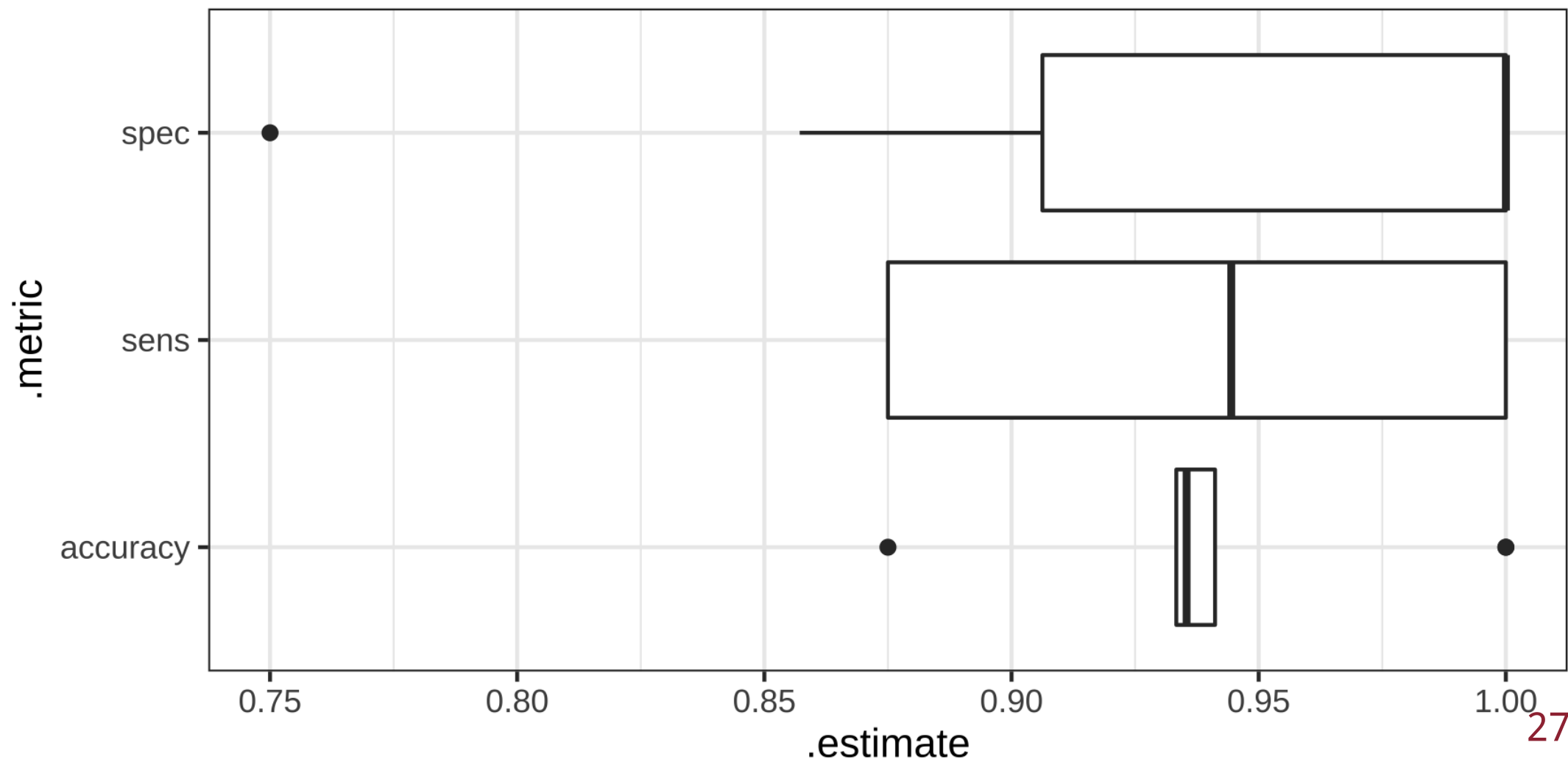
Stratify by **group** to ensure each fold has good representation.

```
resamples <- data %>% vfold_cv(v = 10, strata = Group) (r)
```

```
cv_results <- spec %>%  
  fit_resamples(resamples, metrics = metrics) (r)
```

```
cv_results %>%  
  collect_metrics(summarize = FALSE) %>%  
  ggplot(aes(x = .estimate, y = .metric)) + geom_boxplot()
```

(r)



Cross-validate the decision tree

```
spec <- workflow() %>% add_recipe(  
  recipe(Group ~ ., data = data)) %>%  
  add_model(decision_tree(mode = "classification") %>% set_engine("rpart"))
```

```
cv_results <- spec %>%  
  fit_resamples(resamples, metrics = metrics)
```

```
cv_results %>%  
  collect_metrics(summarize = FALSE) %>%  
  ggplot(aes(x = .estimate, y = .metric)) + geom_boxplot()
```

